

JAY PATEL

Ph.D. Candidate → [\[Google Scholar\]](#) [\[Research Website\]](#) [\[Linkedin\]](#)



Professional Summary

- ❖ 1 Patent under submission, 5 papers accepted: [EMNLP '25](#), [ACL '24](#), [ICWSM '24](#)
- ❖ 5+ yrs. of academic research experience: **AI safety and alignment: NLP (LLM), Computer Vision: Multimodal AI, Computational social science**, Computational game theory (2020-2021).
- ❖ 3+ years of industry work experience: AI researcher, iOS Software Engineer, Full Stack Engineer.
- ❖ [USAToday \(Media Coverage\)](#): Computer vision model to detect [Kekistani flag](#) from 1M 4chan images. [\[Link\]](#)

Industry Experience

AI Research Intern – Comcast, NY

Summer '25

Software Engineer Intern, iOS – IMVU, Redwood City, CA

July 2017 - Dec 2017

Software Engineer Intern – Attala Systems Corp., San Jose, CA

March 2017 - July 2017

iOS Application Developer – Cimcon Software PVT LTD, India

April 2014 - Aug 2015

Conference Publications (🍷 - Submitted | ✅ - Accepted)

🍷 Evaluating Multimedia through Contextual Grounding using VLMs/LLMs – ACL '26

Automating detection of narrative shifts by representing contextual info. for videos

🍷 AI Personas: Evaluating and Steering LLMs to Behave Socially – ACL '26

Exploring persona shift in LLMs for safety by aligning towards social values (e.g., lawful good)

✅ Evaluating LLMs for Social Media Content Moderation – EMNLP '25

Evaluated 8 LLMs to benchmark hate-speech moderation, generated **[3 billion]**+ tokens

✅ Podcast Outcasts: Understanding Rumble's Podcast Dynamics – ACL: NLP4DH '25

Analyzed **[13K hours]**+ of videos from Youtube & Rumble to understand the political leaning

✅ Aya dataset: An ... instruction tuning (Cohere Labs: Open-source contribution) – ACL '24

Lead contributor to process and release **[513 million]**+ data points [\[Huggingface-Aya\]](#)

✅ iDRAMA-rumble-2024: A Dataset of Podcasts from Rumble – AAAI ICWSM '24

Multimedia (videos) data analysis, released **[650K]**+ data points [\[Huggingface-Rumble\]](#)

✅ iDRAMA-scored-2024: A Dataset of the Scored Social Media Platform – AAAI ICWSM '24

Collected, analyzed & released **[56 million]**+ social media posts [\[Huggingface-Scored\]](#)

🍷 From HODL to MOON: Understanding ... Cryptocurrency Ecosystem

Trained RoBERTa model for detecting emotions from **[150 million]**+ Reddit posts

Education

Ph.D. Candidate, Computer Science, State Uni. of NY @Binghamton, NY

Aug 2021 - Present

MS-Research (AI), Computer Science & SE, Auburn University, AL

May 2021

MS, Computer Science, Northwestern Polytechnic University, CA

Dec 2016

BE, Computer Engineering, Gujarat Technological University – India

May 2014

Technical Skills

Languages: (Proficient - Python, Swift, Objective-C, Java) | Intermediate - Ruby, C/C++, JS, Typescript

AI/ML/Python: Pytorch, Tensorflow, Huggingface, LangChain, vLLM, Numpy, Pandas, Matplotlib, NetworkX, Scikit-Learn, Statsmodels, Spacy, cuML, Django, Flask, RESTful & Fast API, Multiprocessing, Celery, PySpark

Models: Transformers, LLM, Vision models, CLIP, SAM. [Model finetuning, training & Mechanistic Interpretability]

Cloud & Databases: AWS, GCP, Docker, MongoDB, Postgresql, Vector-DB (qdrant), RAG, Agentic AI. [Git, LaTeX]

More about Jay...

Research Experience

AI Research Intern – Comcast

- ❖ Developed a computer vision pipeline to automate the localization of important plays in sports videos
 - The pipeline generated output videos got 10% higher rating than the third-party service
- ❖ Extract game-state information using object detection and concurrent multi-point object tracking
 - A novel algorithm to reduce object detection noise by more than 30% for a video
- ❖ Benchmarked VLMs (Vision Lang. Models) in detecting game entities and player actions

Analyzing and evaluating multimedia content through contextual grounding

- ❖ Developed scalable information extraction and knowledge representation system for multimedia data
- ❖ Using CLIP embeddings clustered (UMAP + HDBSCAN) image-frames extracted from 13K+ videos to analyze visual elements/entities used in political discussion
- ❖ Utilized face detection models to construct a personality network for podcast videos
- ❖ Implemented automated detection of narrative shifts using RAG, LLM-as-Judge, and extracted information from videos, like, transcripts, image captioning, etc.

Evaluating LLMs for safety and alignment

- ❖ Developed a prompting technique to increase the LLMs' performance in detecting antisemitism
- ❖ Introduced novel divergence metrics to compare LLM-generated responses
- ❖ Using activation vectors, steering LLMs during inference-time to align the generation towards pre-defined curated personas (steered method expresses the traits with 10% higher than prompting)

Machine learning models for characterizing sentiments on Web

- ❖ Created a synthetic image dataset and trained a DenseNet model to detect hateful symbols (e.g., Nazi flags) in images, achieving 90% AUC-ROC.
 - Scaled model's inference to 1 million images in under 3 hours.
- ❖ Trained a RoBERTa model to detect emotions from Reddit posts, inferring emotions from 152 million crypto-related community posts.

Game-theoretic framework to evolve AI agents

- ❖ Designed a game-theoretic framework facilitating AI agents to play a multiplayer economic game.
- ❖ Implemented attacker and defender AI agents using a competitive coevolutionary algorithm to evolve winning strategies (e.g., greedy agents spend all given money on short-term rewards).
- ❖ Designed an objective function inspired by Claude Shannon's chess game state-eval equation for strategic decision-making.

Leadership, Volunteer, and Organizational Experience

Research Lead – [Binoloop, Canada](#)

Aug 2024 - Aug 2025

Gujarati Language Ambassador – [Cohere Lab's Aya](#)

Aug 2023 - Feb 2024

Founder & Lead Organizer @[GDG Cloud Auburn](#)

March 2019 - March 2021

Media Designer @Indian Students Association, Auburn Uni.

Jan 2019 - Aug 2020

Volunteer tutor to Elementary schools @[Code.org](#) Volunteer

Jan 2017 - Aug 2021

Graduate Teaching Assistant

Jan 2018 - Dec 2022

- ❖ CS515 - Social Media Data Science Pipeline (Fall '22)
- ❖ COMP3500 - Introduction to Operating Systems (Spring '18)
- ❖ COMP3220 - Programming Languages (Spring '18, Fall '18, Fall '19, Spring '20)
- ❖ COMP7270 - Advanced Algorithms (Spring '19)